

NETWORK & SERVER DEPLOYMENT PROJECT

Instructor : Buntha Chay

Subject : System Administrator

Group Members:

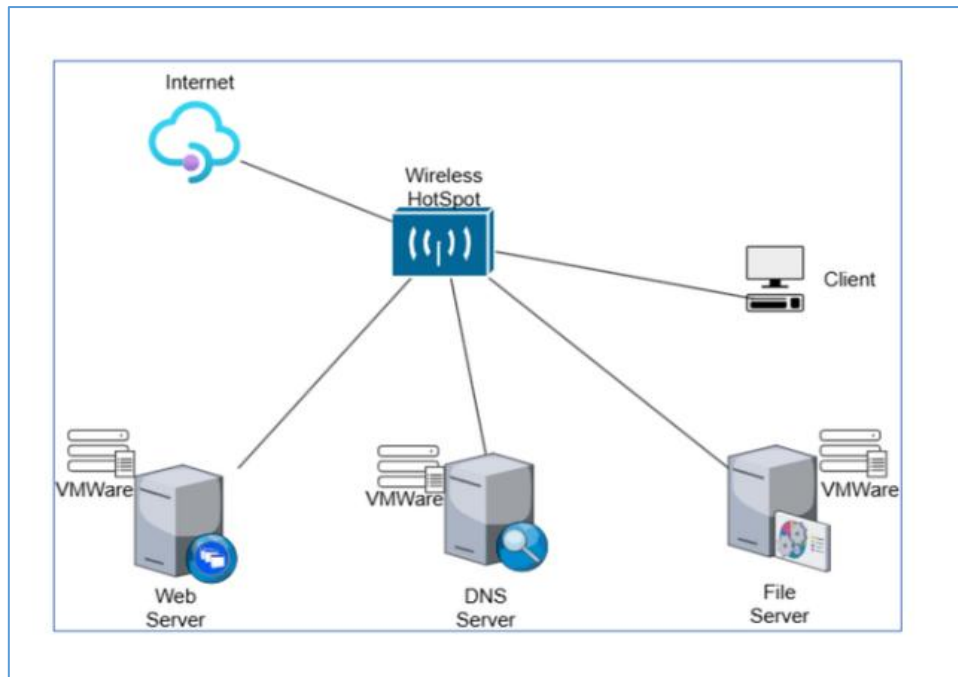
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1. Introduction

This document describes the design, configuration, and testing of a small network built for this course project. The network consists of a Web Server, a DNS Server, and a File Server, all deployed on VMware and connected through a wireless hotspot, with a Client machine used to access and verify each service.

2. Network Design

The diagram below shows the overall network layout: the Client and three servers connect to a shared Wireless HotSpot, which also provides the path to the Internet.



2.1 Component Roles

- Wireless HotSpot — provides connectivity between the Client, all three servers, and the Internet.
- Web Server — hosts the group's website, accessible by IP address and by domain name.
- DNS Server — resolves the chosen domain name to the Web Server's IP address.
- File Server — runs SAMBA and lets the Client upload and download files.
- Client — the machine used to test and demonstrate every service above.

3. IP Addressing Scheme

All devices use static IP addresses within the same subnet, as required. Replace the example values below with the actual addresses your group configured.

Node Name	Role/server	IP Address	Configured Domain
VM1	Web Server(HTTP)	10.11.125.10	mygroup.lab
VM2	DNS Server(BIND9)	10.11.125.20	ns.mygroup.lab
VM3	File Server / Storage	10.11.125.30	files.mygroup.lab
Client Pc	Window Workstation	Dynamic/ Static	Preferred DNS: 10.11.125.20

4. Server Configuration

4.1 Web Server

- Operating system: Ubuntu Server 25.04
- Web server software: HTTP
- Static IP address: 10.11.125.10

4.2 DNS Server

- Operating system and DNS software: BIND9
- Static IP address: 10.11.125.20
- Domain name configured: mygroup.lab
- DNS zone / record details: A record pointing the domain to the Web Server's IP

4.3 File Server (SAMBA)

- Operating system: Ubuntu Server 25.04
- Static IP address: 10.11.125.30
- Permissions: [read/write settings, allowed users]

4.4 SSH / Remote Access

- SSH service enabled on: Web Server, DNS Server, File Server (port 22)
- Client tool used: PuTTY

5. Client Configuration

- PuTTY installed for SSH access to all three servers
- Connected to the same Wireless HotSpot as the servers

6. Testing & Verification

Each requirement below was tested from the Client machine:

- Client and servers reach each other over static IPs in the same range
- SSH remote access works from the Client to every server via PuTTY
- Website loads successfully via the Web Server's IP address
- Website loads successfully via the domain name (through DNS resolution)
- File upload from Client to File Server succeeds
- File download from File Server to Client succeeds

7. Conclusion

Through this project, our group successfully designed, deployed, and tested a small multi-server network on VMware, consisting of a Web Server (Apache2), a DNS Server (BIND9), and a File Server (SAMBAs), all connected via a shared Wireless HotSpot and accessed from a Client machine.

We configured static IP addressing across all devices, set up a custom domain (mygroup.lab) resolved through our own authoritative DNS server, and verified that the website could be reached both by IP address and by domain name. We also configured SAMBA to allow

secure file uploads and downloads between the Client and File Server, and enabled SSH access to all three servers for remote administration via PuTTY.

The project gave us hands-on experience with several real-world networking concepts: how DNS zone files and A records work, how virtual machine networking is affected by adapter modes (NAT, Host-only, and Bridged), and how to diagnose connectivity issues methodically — starting from basic ping tests, through checking IP conflicts, and validating DNS resolution end-to-end. One of the most valuable lessons was troubleshooting a duplicate static IP conflict between two of our VMs, which taught us the importance of verifying network configuration carefully before assuming a service-level problem.

Overall, this project strengthened our understanding of how DNS, web hosting, file sharing, and remote administration work together in a functioning network, and gave us practical experience resolving the kind of issues system administrators encounter in real deployments.